

— PHYSICS 505 – ELECTROMAGNETIC THEORY – SPRING 2026 —

Objective: gain working understanding of the fundamental concepts of classical electromagnetic theory and experience in solving problems at a level appropriate for following research literature and real world applications.

Methodology: we will explore applications of the theory by solving problems after reviewing the basic concepts; we will also spend a few sessions in the computer lab.

Instructor: Yağmur Torun (he, him, his)

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Office hours: TBD by polling.

Textbook: Classical Electrodynamics, 3rd edition, J. D. Jackson, Wiley.

You should read any assigned sections from the book before each lecture and review the material again afterward for best results.

Lecture notes: will be posted online and may include material not covered in class.

[Canvas](#) will be used for posting course documents and grades.

Lectures: are in RE-103 at 13:50-15:05 on Tue/Thu. We may skip some intermediate steps or go over derivations quickly in order to spend more time on the conceptual foundations or computational exploration.

Recitation: if possible, hours to be determined by polling.

Computing: sessions in the department instructional computing lab (PS-112) devoted to demonstrating computational techniques will replace some of the lectures (with advance notice for individual sessions).

Homework: problems will be due most weeks to be submitted in electronic form.

Project: This is a paper review or an application/subject you will study, document in a report and present to the class.

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Grading: In-class performance 10% – Exams 30% – Homework 40% – Project 20%

The lowest entry will be removed when calculating the homework score.

I grade each problem (exam or homework) with an integer score out of 5, you should attempt every problem to take advantage of generous partial credit,

Work submitted	Score
No attempt	0 (no effort, no credit)
Related to the problem but no progress	1 (you thought about it)
Completely wrong and/or substantially incomplete	2 (you tried)
Includes some correct material	3
Mostly good	4
Correct (modulo a sign, typo, numerical mistake)	5

Academic conduct: all students are expected to follow the [Code of Academic Honesty](#) in the student handbook. A first violation results in a grade of zero for that assignment or exam. Subsequent violations result in a failing grade in the course.

ADA Accommodations: in order to receive accommodations, students with documented disabilities should contact the [Center for Disability Resources \(CDR\)](#) as soon as possible. CDR can be reached by phone at 312-567 5744 or email at disabilities@illinoistech.edu and is located in Suite 3F3-1 at the IIT Tower (10 W. 35th St).

Title IX Compliance: Illinois Tech prohibits all sexual harassment, sexual misconduct, and gender discrimination by any member of our community. This includes harassment among students, staff, or faculty. Sexual harassment of a student by a faculty member or sexual harassment of an employee by a supervisor is particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment. Illinois Tech encourages anyone experiencing sexual harassment or sexual misconduct to speak with the Office of Title IX Compliance for information on support options and the resolution process. You can report sexual harassment electronically at iit.edu/incidentreport, which may be completed anonymously. You may additionally report by contacting the Title IX Coordinator, Virginia Foster at foster@illinoistech.edu or the Deputy Title IX Coordinator at eespeland@illinoistech.edu. For confidential support, you may reach Illinois Tech's Confidential Advisor at (773) 907-1062. You can also contact a licensed practitioner in Illinois Tech's Student Health and Wellness Center at student.health@illinoistech.edu or (312) 567-7550. For a comprehensive list of resources regarding counseling services, medical assistance, legal assistance and visa and immigration services, you can visit the Office of Title IX Compliance website at iit.edu/title-ix/resources.
